



Project Title: Investigating the influence of genetic risk factors of PD on peripheral immune cells to identify potential biomarkers for prediction of conversion of non-manifesting to manifesting carriers

PPMI Project ID: 213

Dr Andy Howden, University of Dundee

Summary (or Abstract)

This project aimed to use a combination of flow cytometry, mass spectrometry and single cell sequencing to identify biomarkers for PD in peripheral blood immune cells and plasma. The study has analyzed PBMC and plasma from PPMI which were collected from a carefully selected cohort consisting of LRRK2 mutation carriers with PD, non-manifesting LRRK2 mutation carriers, idiopathic PD and healthy controls (n=20 for each group). The methods described below summarize the analysis of plasma samples by mass spectrometry. Plasma is a rich resource for biomarker discovery but is a challenging sample to analyze due to the large dynamic range of protein abundance within the sample. A small number of proteins make up a large proportion of all protein within the samples, making the detection and quantification of low abundance proteins a challenge. Using high sensitivity mass spectrometry we have identified and quantified over 1000 proteins in un-depleted plasma and over 2000 proteins in plasma with the top 14 most abundant proteins depleted.

Method

Processing plasma for mass spectrometry

Plasma samples were processed by two approaches: 1) un-depleted plasma 2) plasma with the top 14 most abundant proteins depleted. Using the un-depleted approach, 5µl of cryopreserved plasma was transferred to a 1.5ml lo-bind Eppendorf tube and 45µl SDS lysis buffer added for a final concentration of 5% SDS, 50mM TEAB and 10mM TCEP. Samples were incubated at 95°C for 5 minutes and then allowed to cool to room temperature. Iodoacetamide was added to each sample to a final concentration of 20mM and samples incubated in the dark for 1 hour. Protein quantity was measured using the EZQ assay (Thermo). For each sample 150µg of protein was loaded onto a S-trap multiwell plate following the manufacturers instructions (Protifi). Samples were digested with the addition of 10µg trypsin for 2 hours at 37°C and then eluted and dried by speed-vac. Samples were re-suspended in 75µl of 1% formic acid and peptides quantified using the CBQCA peptide quantification assay (Thermo). For mass spectrometry analysis, 1.5µg of peptide was suspended in a final volume of 11µl 1% formic acid. Samples were analyzed by data independent acquisition mass spectrometry as described below.





Mass spectrometry data acquisition

Analysis of peptides was performed on a Q Exactive HF, Mass Spectrometer (Thermo Scientific) coupled to a Dionex Ultimate 3000 RS (Thermo Scientific). LC buffers used are the following: buffer A (0.1% formic acid in Milli-Q water (v/v), buffer B (80% acetonitrile and 0.1% formic acid in Milli-Q water (v/v), loading buffer (0.1% TF in Milli-Q). Aliquots of 11 μ l of each sample were loaded at 10 μ L/min onto a trap column (100 μ m $\times$ 2 cm, PepMap nanoViper C18 column, 5 μ m, 100 Å, Thermo Scientific) equilibrated in 0.1% TFA. The trapping column was washed for 6 min at the same flow rate with 0.1% TFA and then switched in-line with a Pharma Fluidics, 200 cm, μ PAC nanoLC C18 column. The column was equilibrated at a flow rate of 300nl/min for 30 min. The peptides were eluted from the column at a constant flow rate of 300 nl/min with a linear gradient from 1% buffer B to 3.8% buffer B in 6 min, from 3.8% B to 12.5% buffer B in 40 min, from 12.5% buffer B to 41.3% buffer B within 176 min and then from 41.3 % buffer B to 61.3% buffer B in 14 min. The gradient is finally increased from 61.3% buffer B to 100% buffer B in 1 min, and the column was then washed at 100% buffer B for 10 min. Two blanks were run between each sample to reduce carry-over. The column was kept at a constant temperature of 50oC.

Q-exactive HF was operated positive ionization mode using an easy spray source. The source voltage was set to 2.3 Kv and the capillary temperature was 250oC. Data were acquired in Data Independent Acquisition Mode as previously described (Doellinger et al., 2020), with small modifications. A scan cycle comprised a full MS scan (m/z range from 345-1155), resolution was set to 60,000, AGC target 3×10^6 , maximum injection time 200 ms. MS survey scans were followed by DIA scans of dynamic window widths with an overlap of 0.5 Th. DIA spectra were recorded at a resolution of 30.000 at 200 m/z using an automatic gain control target of 3×10^6 , a maximum injection time of 55 ms and a first fixed mass of 200 m/z. Normalised collision energy was set to 25 % with a default charge state set at 3. Data for both MS scan and MS/MS DIA scan events were acquired in profile mode.

Data analysis

For all samples, raw mass spec data was analyzed using the spectronaut analysis software (Biognosys, version 17.5.230413.55965) using a hybrid library approach. The hybrid library was generated from fractionated plasma samples (analyzed by data dependent acquisition) along with experimental samples (analyzed by data independent acquisition). The following search parameters were used: the FDR threshold was set to 1% Q-value at both the Precursor and Protein level, protein LFQ method was set as Quant 2.0. The data were searched against the human SwissProt database with isoforms (June 2022).





Processing depleted plasma samples

For depleted plasma, 10µl of cryopreserved plasma was loaded onto a high-select Top14 column following the manufactures instructions (Thermo). After addition of plasma, columns were gently, shaken at room temperature for 10 minutes ensuring that the sample is fully mixed with the column resin. After incubation, proteins were eluted by centrifugation at 1000g for 2 minutes. The eluted protein sample was then processed as described above for the un-depleted approach.

References

Doellinger, J., Blumenscheit, C., Schneider, A. and Lasch, P. Isolation Window Optimization of Data-Independent Acquisition Using Predicted Libraries for Deep and Accurate Proteome Profiling. *Analytical Chem* (2020) 92(18):12185-12192. doi: 10.1021/acs.analchem.0c00994.

About the Authors

This document was prepared by Dr Andy Howden, Division of Cell Signalling and Immunology, University of Dundee. For more information please contact Andy at +44 1382 385828 or by email at a.howden@dundee.ac.uk.

Notice: This document is presented by the author(s) as a service to PPMI data users. However, users should be aware that no formal review process has vetted this document and that PPMI cannot guarantee the accuracy or utility of this document.

